

Supplementary Materials to “Single Antenna Bio-Sensing for Non-invasive Respiratory and Cardiac Activity Monitoring”

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A. EFFECTS OF WEARING LAYERS OVER ANTENNA SENSOR

Wearing additional layers over the antenna sensor alters the material composition in the near field of the antenna. This change slightly modifies the effective permittivity seen by the antenna, resulting in a small shift in its impedance at the operating frequency and subsequently affecting the baseline of the antenna's S_{11} measurement. However, vital signals can still be retrieved after this baseline readjustment. It should be noted that this effect will be minimal for antenna sensors whose near-field patterns are focused toward the body, such as patch antennas, where the backside is shielded by a ground plane, preventing exposure to external materials and free space.

Figure 1 illustrates a series of experiments where an extra layer of clothing was added over the sinusoidal dipole antenna sensor during operation. Each experiment begins with the patient wearing the sensor over the chest, with a thin T-shirt between the body and the sensor. The shaded areas indicate when the additional wearable was put on over the antenna, which caused artifacts due to vigorous movements of the volunteer during the dressing process. The subsequent epochs show signals with a shifted baseline due to the added layer on top. As demonstrated, despite the baseline shift, vital signals remain detectable and can be extracted from the raw data.

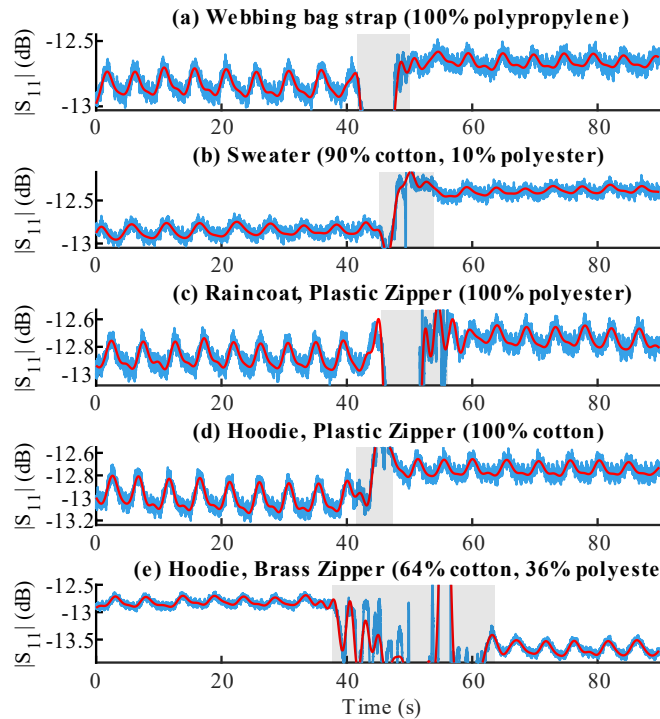


Figure 1 Sample raw signals recorded before and after wearing extra clothing layer over the antenna sensor. Gray regions show the dressing duration. Red trace shows the filtered BR signal.